AI DRIVEN PLANT HEALTH MONITORING AND TARGETED CHEMICAL SPRAYING MECHANISM

AN INTERNSHIP REPORT

Submitted by

KAVIN SANTHOSH S 811722106037

in partial fulfillment of the requirements for the award degree of

Bachelor in Engineering

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY

(AUTONOMOUS)

SAMAYAPURAM - 621112

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BONAFIDE CERTIFICATE

This is to certify that the work embodies in this dissertation entitled "AI DRIVEN PLANT HEALTH MONITORING TARGETED CHEMICAL SPRAYING MECHANISH" being submitted by "KAVIN SANTHOSH S", "811722106037" for partial fulfilment of the requirement for the award of "Certificate" during the period 2/02/2026 to 2/03/2026 internship programme is a record of a Bonafide piece of work, which was carried out by him under my supervision and guidance in the Department of Electronics And Communication Engineering of K.Ramakrishnan College of Technology (Autonomous).

Date of Viva Voce: ……

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INTERNAL EXAMNIER

ABSTRACT

Plant diseases significantly affect agricultural productivity, leading to major economic losses for farmers. Traditional detection methods rely on manual inspection, which is time-consuming and inaccurate. This project presents an AI-driven plant health monitoring and targeted chemical spraying system using deep learning and embedded hardware. Plant leaf images are captured by a camera and analyzed using the U-Net algorithm to detect infected regions based on discoloration, spots, and texture changes. Upon detection, a control signal is sent to the ESP8266 microcontroller, which activates a relay-driven DC pump to spray pesticide on the affected plant. The system achieved classification confidence values of 90–95%, reducing chemical usage, manual labor, and environmental pollution.

Keywords : Plant disease detection, Deep learning, U-Net algorithm, ESP8266, Automated pesticide spraying, Precision agriculture, Smart farming, OpenCV.

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I am deeply indebted to our internship coordinator, **Mrs. R. Bhavani, M.E., MBA., (Ph.D).,** Assistant Professor, Department of Electronics and Communication Engineering, for their valuable suggestions and support in successfully completing the internship.

Finally, we thank our parents, friends, and well-wishers for their kind support.

SIGNATURE

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CERTIFICATE OF INTERNSHIP

This is to certify that **Mr. KAVIN SANTHOSH S (811722106037)**, a student of Final Year Electronics and Communication Engineering of 2022–26 Batch at K. Ramakrishnan College of Technology (Autonomous), has successfully completed an Internal Internship under Project Based Internship Category on **"AI DRIVEN PLANT HEALTH MONITORING TARGETED CHEMICAL SPRAYING MECHANISH"** in the Department of Electronics and Communication Engineering from 2/02/2026 to 2/03/2026.

During the internship period, the student actively done the following task:

- Studied and analyzed latency issues in real-time applications Implemented edge computing techniques to reduce communication delay

The internship was aimed at enhancing practical skills, technical knowledge and problem-solving abilities aligned with the academic curriculum.

We appreciate the commitment and enthusiasm shown by the student during the internship.

We wish all the best for her future endeavors.

Date: With Regards,

Place: Samayapuram, Tiruchirappall

TABLE OF CONTENTS

CHAPTER NO.	TITLE	PAGE NO.
	ABSTRACT	**iii**
	INTERNSHIP CERTIFICATE	**v**
	LIST OF FIGURES	**vii**
	LIST OF SYMBOLS AND ABBREVIATIONS	**viii**
1	**INTRODUCTION**	**1**
	1.1 COMPANY PROFILE AND ITS INDUSTRIES	1
	1.2 AREA OF INTERNSHIP (IMAGE PROCESSING)	1
	1.3 PROBLEM STATEMENT	2
	1.4 OBJECTIVE OF THE INTERNSHIP	3
2	**EXECUTIVE SUMMARY**	**4**
3	**ACTIVITIES**	**7**
	3.1 PROJECTS/MODULES UNDERTAKEN	7
	3.2 TOOLS AND TECHNOLOGIES USED	9
	3.3 SKILLS ACQUIRED	10
	3.3.1 Professional Skills	10
	3.3.2 Technical Skills	10
	3.3.3 Personal Growth	10
4	**CHALLENGES AND SOLUTIONS**	**11**
5	**CONCLUSION**	**13**
	REFERENCES	**14**
	STUDENT FEEDBACK	
	FEEDBACK FROM INDUSTRY	

LIST OF FIGURES

FIGURE NO.	TITLE	PAGE NO.
2.1	Block Diagram	4
2.2	SnapShot Of ProtoType Model	5

LIST OF SYMBOLS AND ABBREVIATIONS

AI	-	Artificial Intelligence
CNN	-	Convolutional Neural Network
CPU	-	Central Processing Unit
DC	-	Direct Current
GPIO	-	General Purpose Input/Output
GND	-	Ground
IDE	-	Integrated Development Environment
IOT	-	internet of Things
KNN	-	K-Nearest Neighbor
LED	-	Light Emitting Diode
MHz	-	Megahertz
ML	-	Machine Learning
OpenCV	-	Open Source Computer Vision Library
OS	-	Operating System
PC	-	Personal Computer
RAM	-	Random Access Memory
RTS	-	Request to Send
RX	-	Receive
SD	-	Secure Digital
SVM	-	Support Vector Machine
TX	-	Transmit
UART	-	Universal Asynchronous Receiver Transmitter
USB	-	Universal Serial Bus
VCC	-	Voltage Common Collector
Wi-Fi	-	Wireless Fidelity

CHAPTER 1

INTRODUCTION

1.1 COMPANY PROFILE AND ITS INDUSTRIES

K Ramakrishnan College of Technology (KRCT) is an autonomous engineering institution affiliated with Anna University. The institution is well known for its strong emphasis on research, innovation, and practical learning.The Department of Electronics and Communication Engineering actively contributes to research in areas such as embedded systems, IoT, artificial intelligence, and automation technologies. The department provides facilities such as microcontroller development kits, IoT modules, and software tools, which supported the development of the AI-based plant health monitoring system during the internship.

1.2 AREA OF INTERNSHIP (IMAGE PROCESSING)

The internship was conducted in the domain of Artificial Intelligence–based Smart Agriculture Systems, with a specific focus on plant disease detection and automated targeted chemical spraying using intelligent technologies. The primary research area involved the study and application of deep learning and computer vision techniques for identifying plant health conditions from leaf images and enabling precision-based pesticide application. The work mainly focused on the use of Deep Learning models such as Convolutional Neural Networks (CNN) and U-Net architectures to analyze plant leaf images and detect disease symptoms like spots, discoloration, and texture variations. This approach helps in early detection of plant diseases and improves crop protection efficiency.

1.3 PROBLEM STATEMENT

Agriculture faces significant challenges due to plant diseases, which directly affect crop quality, productivity, and farmer income. Early detection of plant diseases is difficult using traditional farming methods because most farmers rely on manual observation. This process is time-consuming, less accurate, and not suitable for monitoring large agricultural fields.

One of the major problems in conventional farming is the excessive use of pesticides due to uniform spraying practices. Farmers often spray chemicals across the entire field without identifying the exact infected areas. This leads to chemical wastage, increased production cost, soil degradation, and environmental pollution.

Another important issue is the lack of real-time plant health monitoring systems. Existing methods do not continuously track plant conditions or provide early warnings about disease symptoms.

Furthermore, manual pesticide spraying methods require significant labor effort and time. In large farms, it becomes difficult to inspect every plant and apply treatment precisely

Therefore, there is a need for an intelligent and automated system that can continuously monitor plant health, detect diseases at an early stage, and automatically apply pesticides only where required. Such a system can reduce chemical usage, improve crop protection efficiency, minimize human effort, and support sustainable precision agriculture.

1.4 OBJECTIVES OF THE INTERNSHIP

The primary objective of the internship was to gain practical knowledge by working on a real-time smart agriculture project and understanding how artificial intelligence and embedded systems can be applied to solve real-world farming problems.

The specific goals of the internship include:

1. To design and develop an AI-based plant health monitoring system for early disease detection
2. To implement image processing and deep learning techniques for identifying plant diseases
3. To integrate camera modules, ESP8266 microcontroller, relay circuits, and pump mechanisms
4. To develop skills in AI model training, OpenCV image analysis, and hardware automation
5. To understand system architecture, hardware–software integration, testing, and debugging processes
6. To implement automated targeted chemical spraying to reduce pesticide wastage
7. To enhance technical skills in Python, embedded systems, and smart agriculture technologies
8. To improve problem-solving ability and practical engineering knowledge in precision farming systems
9. To achieve real-time plant monitoring and automated decision-making using Artificial Intelligence
10. To develop a cost-effective and environmentally friendly solution for sustainable agriculture practices

CHAPTER 2

EXECUTIVE SUMMARY

During the internship period, I worked on the project titled "AI Driven Plant Health Monitoring and Targeted Chemical Spraying Mechanism," which focuses on improving agricultural efficiency using modern technologies. The primary objective of the internship was to develop an intelligent system capable of detecting plant diseases and performing automated pesticide spraying.

The internship involved the development of an intelligent smart agriculture system that uses artificial intelligence and embedded systems to monitor plant health and automate pesticide application. The primary objective was to design, implement, and evaluate a system capable of detecting plant diseases from leaf images and activating a targeted spraying mechanism to treat infected plants.

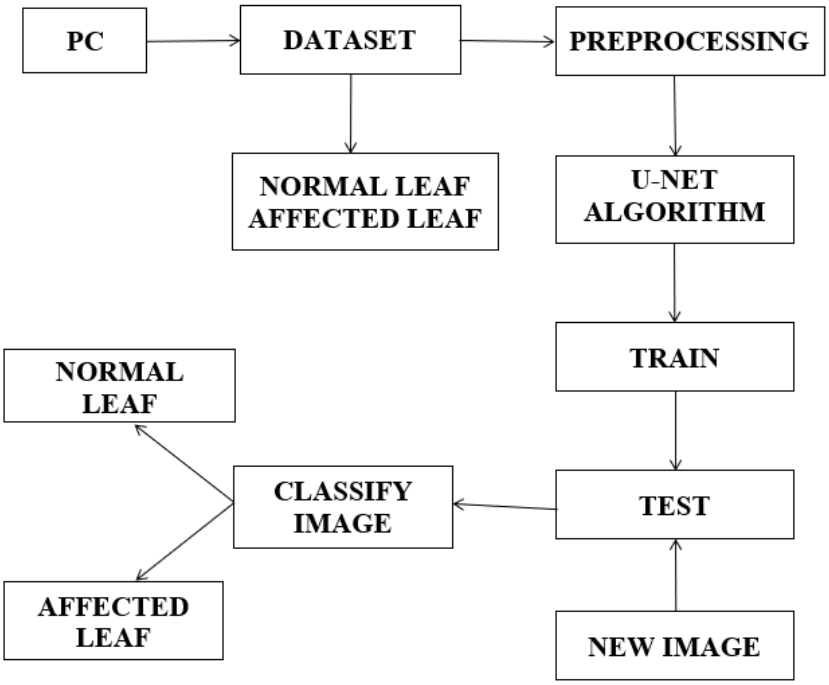

Figure 2.1 Block Diagram

The work included multiple stages such as dataset collection, image preprocessing, AI model development, hardware interfacing, and system testing. A deep learning model based on Convolutional Neural Networks was used for plant disease detection. The system was implemented using Python programming and image processing techniques with OpenCV.

Tasks and Responsibilities Assigned:

Literature Review:

Conducted a detailed study on plant disease detection methods, deep learning techniques, and smart agriculture systems to build a strong theoretical understanding.

System Design:

Designed the overall architecture of the plant health monitoring system, including image acquisition, disease detection, and automated spraying modules.

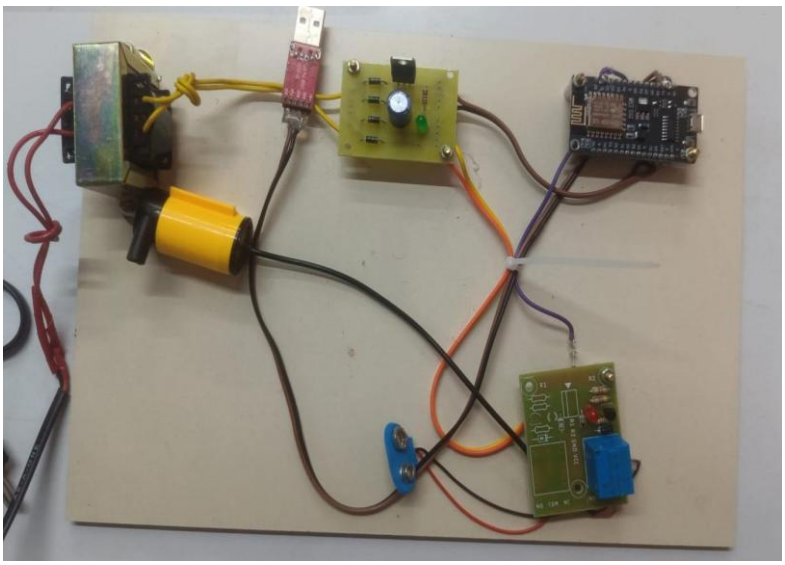

Figure 2.2 Snapshot Of Prototype Model

AI Model Development:

Developed and trained deep learning models for classifying plant leaves as healthy or diseased using labeled image datasets.

Image Processing:

Implemented preprocessing techniques such as image resizing, filtering, and feature extraction to improve model accuracy.

Hardward Integration:

Interfaced ESP8266 microcontroller with relay modules and DC pump to enable automated pesticide spraying based on detection results.

Real-Time Monitoring:

Developed a system for continuous plant monitoring using a camera module and automated decision-making process.

System Testing and Validation:

Tested the system with different plant samples and verified detection accuracy and spraying response.

Debugging and Optimization:

Identified and resolved errors in both software and hardware to improve system efficiency and reliability.

Report Preparation:

The internship provided valuable hands-on experience by combining theoretical knowledge with practical implementation in artificial intelligence and embedded systems. It enabled a deeper understanding of smart agriculture technologies and real-time automation systems. Working closely with the project guide helped in gaining insight into modern AI applications in agriculture and sustainable farming practices.

CHAPTER 3

ACTIVITIES

3.1 PROJECTS / MODULES UNDERTAKEN

The primary project undertaken during this internship was the design and development of an intelligent system for monitoring plant health and performing automated targeted pesticide spraying using artificial intelligence and embedded systems. The project consists of multiple modules, each addressing a specific function in the smart agriculture system.

Module 1: Plant Disease Detection Using AI

This module focuses on detecting plant diseases using image processing and deep learning techniques. A camera module is used to capture images of plant leaves, which are then processed and analyzed using AI algorithms.

The system uses Convolutional Neural Networks to classify leaves as healthy or diseased. Image preprocessing steps such as resizing, filtering, and normalization are performed using OpenCV to improve detection accuracy.

The trained model identifies disease symptoms such as leaf spots, discoloration, and abnormal patterns. Real-time detection was validated using different plant samples, ensuring reliable classification results.

Module 2: Image Processing and Feature Analysis

This module handles preprocessing and feature extraction from captured plant images. The goal is to enhance image quality and extract relevant features for accurate classification.

Various image processing techniques were implemented, including noise removal, contrast enhancement, and segmentation of infected regions. Feature extraction helps the AI model to distinguish between healthy and diseased leaves more effectively.This module plays a crucial role in improving the performance and accuracy of the disease detection system.

Module 3: Automated Targeted Spraying Mechanism

This module is responsible for activating the pesticide spraying system based on disease detection results. The AI model output is sent to the ESP8266 microcontroller, which controls the spraying mechanism.

A relay driver circuit is used to control a DC pump. When a diseased plant is detected, the relay is activated, and the pump sprays chemicals only on the infected plant. This targeted spraying approach reduces pesticide wastage and minimizes environmental impact.

Module 4: Hardware Integration and Control System

This module focuses on integrating hardware components such as ESP8266, relay modules, power supply, and pump system.

Communication between the AI system and microcontroller is established using serial communication. The microcontroller processes the input signals and controls the spraying mechanism in real time.

The hardware setup was tested to ensure proper synchronization between detection and spraying operations.

Module 5: Real-Time Monitoring and Decision Making

This module enables continuous monitoring of plant health conditions. The system operates in a loop where images are captured, processed, and analyzed periodically.

Based on the AI model output, automated decisions are made without human intervention. This improves efficiency and ensures timely action in case of disease detection.

3.2 TOOLS AND TECHNOLOGIES USED

Hardware Components

1) ESP8266 Microcontroller
2) Camera Module
3) Relay Driver Circuit
4) DC Water Pump
5) Power Supply Unit

Software Tools

Python Used for AI model development and system control OpenCV Used for image processing

TensorFlow/Keras Used for deep learning model training Arduino IDE –Used for programming ESP8266

Technologies

1) Artificial Intelligence
2) Image Processing
3) Embedded Systems
4) IoT-Based Automation
5) Precision Agriculture

3.3 SKILLS ACQUIRED

3.3.1 Professional Skills

Developed knowledge in project planning, execution, and documentation
• Improved teamwork and communication skills during project development
• Gained experience in research and understanding modern smart agriculture technologies
• Learned to approach real-world problems with structured solutions

3.3.2 Technical Skills

•Practical experience in AI model development and training
• Proficiency in Python programming and OpenCV-based image processing
•Hands-on experience in microcontroller programming and hardware interfacing
•Understanding of real-time system integration and automation
• Knowledge of precision agriculture systems and IoT-based solutions

3.3.3 Personal Growth

This internship significantly improved my confidence in handling real-time engineering projects. I developed strong problem-solving skills by overcoming challenges in AI model accuracy and hardware integration. The experience enhanced my analytical thinking and ability to work independently.

Regular guidance from the project mentor helped me understand the importance of accuracy, testing, and continuous improvement. Overall, the internship strengthened my interest in artificial intelligence, embedded systems, and smart agriculture technologies, motivating me to pursue further learning and innovation in this field.

CHAPTER 4

CHALLENGES AND SOLUTIONS

During the internship focused on developing an AI-Driven Plant Health Monitoring and Targeted Chemical Spraying Mechanism, multiple challenges were faced due to the complex intersection of AI software development and embedded hardware integration in a dynamic agricultural setting. One significant challenge was ensuring accurate plant disease detection, as the system depended heavily on image processing. Factors such as varying lighting conditions, background noise, and leaf orientation occasionally compromised detection accuracy, making it difficult to achieve consistent results across diverse environmentalsituations.

Another pressing issue was related to the quality of the dataset and the training of the AI model. The system required a substantial number of images representing both healthy and diseased plants to train the model effectively. Limitations in dataset size and inconsistencies in disease manifestation negatively impacted the model's classification performance. Hardware integration presented further complications; components including the ESP8266 microcontroller, relay modules, DC pumps, and power supply needed meticulous connections. Problems such as faulty wiring, unstable power sources, and communication lags at times diminished the system's efficacy.

Additionally, establishing a reliable communication link between the Python-based AI model and the ESP8266 microcontroller proved challenging in the initial phases of the project. Serial communication errors and delays hindered the real-time functionality of the spraying system. Precision in chemical spraying also posed a challenge; ensuring that the pump only activated under specific conditions required careful programming and rigorous testing. Problems such as faulty wiring, unstable power sources, and communication lags at times diminished the system's efficacy.

Testing the system under genuine operational conditions was arduous, necessitating the analysis of multiple plant samples to confirm the accuracy of diagnosis and the corresponding spraying response. Continuous monitoring and iterative testing were essential for validating the performance of the system.

To address these hurdles, a range of effective solutions and strategies was implemented throughout the internship. To enhance the accuracy of disease detection, various image preprocessing techniques—such as resizing, normalization, and noise reduction—were utilized, with tests conducted across different lighting environments to bolster model reliability. Addressing dataset limitations involved augmenting the initial image repository through the collection of additional plant leaf images and techniques like rotation and flipping, which enriched the training process and improved the classification accuracy.

Hardware issues were tackled by systematically organizing connections and ensuring that a stable power supply was maintained for all components. Conducting incremental hardware tests helped pinpoint wiring discrepancies and enhance overall system reliability. Solutions to facilitate robust communication between the AI system and the microcontroller included employing dependable serial communication through PySerial, as well as establishing effective error-handling protocols in the coding, resulting in improved real-time responsiveness of the training process to spraying AI system mechanism.In targeted areas.

Finally, to tighten control over the spraying mechanism, conditions were programmed to ensure the pump activated only when disease detection confidence exceeded a predetermined threshold. This adjustment minimized unnecessary chemical applications, thereby optimizing resource use and reducing environmental impacts

CHAPTER 5

CONCLUSION

The internship successfully achieved its objectives by developing a system capable of detecting plant diseases using artificial intelligence and performing automated targeted pesticide spraying.. Image processing techniques implemented using OpenCV improved the detection accuracy by analyzing leaf patterns, color variations, and disease symptoms.

The integration of AI software with embedded hardware components such as ESP8266, relay modules, and DC pump enabled real-time monitoring and automated decision-making. Testing and evaluation of the system confirmed that it performs reliable disease detection with minimal response time. The targeted spraying mechanism significantly reduced chemical wastage compared to traditional uniform spraying methods.

In conclusion, the internship demonstrated that artificial intelligence and embedded systems can be effectively integrated to develop smart agricultural solutions. The proposed system provides a cost-effective, scalable, and environmentally friendly approach to plant health monitoring and crop protection, laying a strong foundation for future advancements in intelligent farming technologies.

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